Diving into Underwater: Segment Anything Model Guided Underwater Salient Instance Segmentation and A Large-scale Dataset

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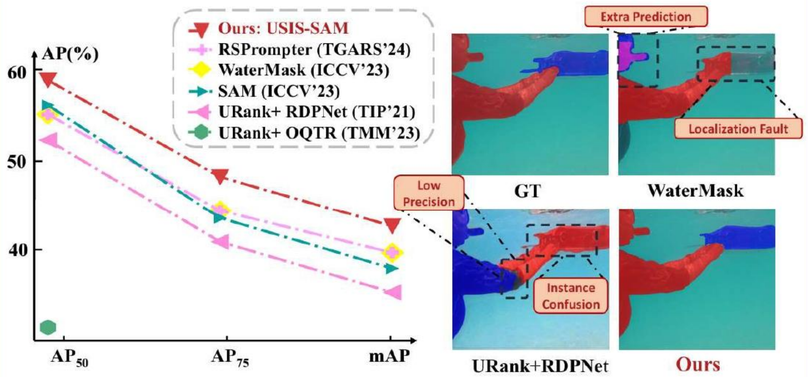
Paper Code

1. Motivation and Challenges



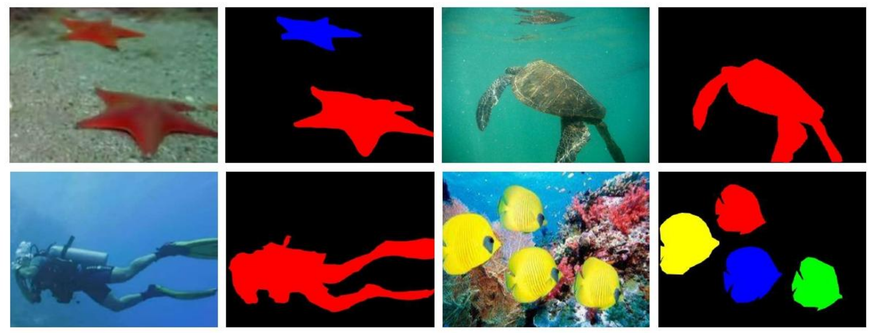
- On the one hand, there is **no general underwater salient image instance segmentation dataset** to promote training and evaluation of the underwater salient instance segmentation (USIS) models.
- On the other hand, even state-of-the-art SIS models trained on large-scale land-based datasets coupled with the best underwater image enhancement algorithms **can't achieve satisfactory performance in underwater environments.**

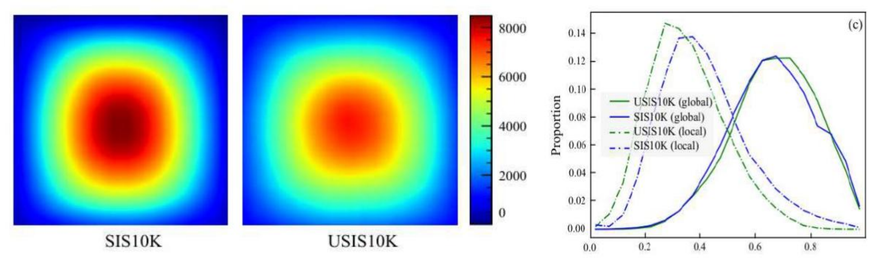
2. Contributions

- We construct the **first large-scale underwater salient instance segmentation (USIS) dataset, USIS10K**, aiming to promote the development of salient instance segmentation for underwater tasks.
- We propose **USIS-SAM**, aiming to **improve the segmentation accuracy in complex underwater scenes.**

3. USIS10K Dataset

We construct the **USIS10K** for the underwater salient instance segmentation task, which contains **10,632 images and pixel-level annotations of 7 categories**. As far as we know, this is **the largest salient instance segmentation dataset**, and includes Class-Agnostic and Multi-Class labels simultaneously. Some examples are given below.

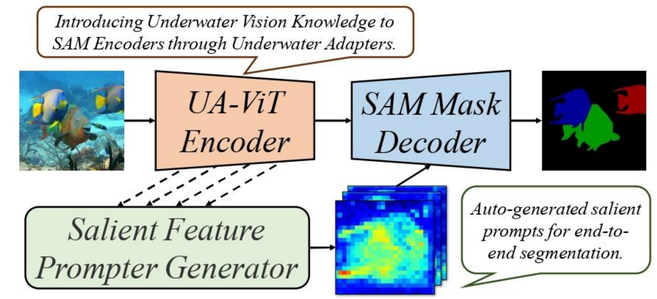




(a) Location of Salient Objects (Less central bias). In the SIS10k dataset (*TMM'23*), 13.5% of the locations have fewer than 1000 instances, while in our dataset, only 2.75% of the locations have fewer than 1,000 instances.

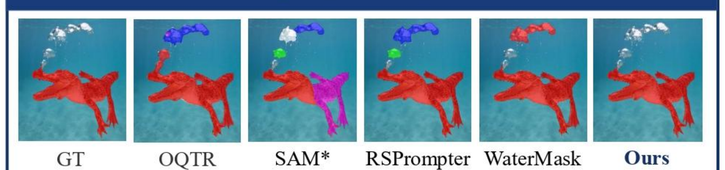
(b) Color Contrast of Salient Instance. Saliency is usually related to the global contrast of the foreground and background. It can be seen that USIS10K has better global contrast and more challenging local contrast.

4. USIS-SAM



- We devise an **Underwater Adaptive Visual Transformer (UA-ViT) Encoder** to incorporate underwater domain visual prompts into the segmentation network.
- We further design an out-of-the-box **Salient Feature Prompter Generator** to auto-generate salient prompters instead of explicitly providing foreground points or boxes as prompts in SAM

5. Result



We retrained/fine-tuned USIS-SAM, the SIS method, the SAM method, and the underwater instance segmentation method on USIS10K. During training, the original parametersof SAM module were frozen for our USIS-SAM and other compared methods. The results show that USIS-SAM leads in all metrics.

**SAM uses output from Faster RCNN as prompts.*